

Beginner	Recognize bias types	Memorize 4 core categories + 10 red flags	Map biases to lifecycle stages 1-5
Intermediate	Predict & prevent	Design studies to mitigate bias	Implement Stage 1-5 checklists in real projects
Advanced	Quantify bias magnitude	Sensitivity analysis, post-stratification, weighting	Measure residual bias after mitigation
Expert	Communicate limitations	Stakeholder education, transparent reporting	Build organizational bias-aware culture

"Your retail client wants to launch a new loyalty program. They believe longer checkout times cause customer dissatisfaction. They ask you to survey customers and propose solutions. What biases would you worry about at each stage, and how would you prevent them?"

"You notice that in your survey data, customers aged 65+ are severely underrepresented (only 5% of respondents vs 25% in population). What type of bias is this, and how would you correct for it?"

"Your boss wants you to publish a report showing how much sales improved after a marketing campaign. You see some improvements but also some metrics declined. The boss only wants the positive results published. How do you handle this?"

Stage	Primary Biases	Prevention Tools	Red Flags	
1. Problem Definition	Confirmation, Anchoring, Availability	Diverse stakeholders; open framing	Too-narrow hypothesis; no alternative considerations	
2. Data Collection	Selection, Exclusion, Volunteer, Instrument, Observer, Response	Probabilistic sampling; multi-mode collection; training	Low response rates; demographic mismatches; leading questions	
3. Data Analysis	Confirmation, Analysis, P-hacking	Pre-registration; blind analysis; multiple testing corrections	p-values clustered near 0.05; single model specification	
4. Result Sharing	Reporting, Framing, Publication, Overconfidence	Balanced presentation; uncertainty ranges; limitations disclosure	Only positive results; no confidence intervals; definitive language	
5. Evaluation	Evaluation, Outcome, Hindsight, Attribution	Pre-defined criteria; control groups; multi-factor assessment	Cherry-picked success metrics; no counterfactuals; attribution to single cause	

